

WEEK 6 - RESEARCH REPORT

Investigating Phishing and Online Financial Fraud

A Pakistan-Focused Review of Research Published Since 2020

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Scope: phishing, social engineering, online banking/payment fraud, open-source defensive practice, and Pakistan-specific challenges.

Abstract

Phishing remains one of the easiest ways for criminals to turn a technical weakness into direct financial loss. Since 2020, research has moved from simple URL blacklists and hand-written rules toward machine learning, deep learning, visual brand recognition and, more recently, multimodal large language models. At the same time, financial-fraud research has increasingly focused on anomaly detection, customer-behaviour modelling and risk-aware decision making. This report reviews recent work published from 2020 onward and connects it with the Pakistani setting. The review finds that modern detectors can achieve strong laboratory results, but operational problems remain: phishing pages change quickly, labelled fraud data are rare, false positives are expensive, and attackers increasingly use social engineering across email, SMS, messaging apps and phone calls. In Pakistan, these issues matter because digital banking and online payments continue to expand while customer trust is directly affected by phishing, identity theft and hacking. The report also discusses practical open-source tools such as Phishpedia, MISP, YARA and Zeek, and ends with three research gaps: local multilingual datasets, adaptive detection against AI-assisted scams, and better cross-organization sharing of fraud evidence.

Keywords: phishing, online financial fraud, digital banking, Pakistan, machine learning, open-source security, threat intelligence

1. Review Method and Scope

The review followed the assignment requirement that every academic paper must be dated 2020 or later. Search terms were built around combinations of phishing, email phishing, phishing webpage detection, online payment fraud, banking fraud, machine learning, deep learning, Pakistan and e-banking. Open-access versions were prioritized from USENIX, arXiv, DOAJ/PMC and publisher repositories. The final reading set combined broad surveys with recent technical systems and Pakistan-focused work. Zotero can be used to store the papers, remove duplicates and export IEEE references; this is useful because phishing research often appears in both preprint and final published versions. The aim was not to list every paper but to identify the main direction of current research and the gaps that still matter in practice.

2. Current Threat Picture

Phishing is no longer limited to a badly written email asking for a password. Modern campaigns may copy the visual identity of a bank, delivery company, government service or social-media platform, and then move the victim to a fake login page or a payment step. Attackers also combine channels: an SMS or WhatsApp message may create urgency, a phone call may imitate bank support, and a fake webpage may finally collect credentials or an OTP. A 2021 systematic review grouped common techniques around spear phishing, email spoofing/manipulation and phone phishing, while also finding machine-learning approaches to be the dominant technical direction in anti-phishing research [1]. During the COVID-19 period, attackers rapidly adapted themes to public fear and uncertainty, showing how quickly social-engineering content can change [2].

This matters for online financial fraud because credential theft is often only the first stage. Once an account is taken over, the defender must identify abnormal behaviour quickly enough to stop a transfer without blocking too many legitimate customers. Financial-fraud research therefore treats detection as both a classification problem and a risk-management problem. Reviews published in 2022 show heavy use

of supervised methods such as support-vector machines and neural networks, but also note growing interest in anomaly detection and unsupervised learning because confirmed fraud cases are rare [6], [7].

3. State of the Art Since 2020

Direction	Example recent work	Strength	Practical limitation
Email/text ML	Butt et al. [5]	Fast classification using email content and features	Can overfit to known datasets and wording
Deep learning	Catal et al. [4]	Learns complex patterns from URLs/content/email	Needs representative data; difficult to explain
Visual brand detection	Phishpedia [3]	Identifies copied brands without phishing training samples	Reference/brand coverage still matters
Intent + interaction	PhishIntention [8]	Checks both visual brand intent and credential-taking behaviour	More computation and browser interaction
Fraud anomaly detection	Ali et al.; Hilal et al. [6], [7]	Useful when fraud labels are limited	Class imbalance and concept drift
Multimodal LLMs	Lee et al. [10]	Combines page appearance, brand reasoning and domain checks	Cost, prompt/model drift, adversarial manipulation

Visual phishing detection. Phishpedia is an important 2021 example because it recognizes identity logos and matches logo variants, producing an explanation of the likely targeted brand. In its field deployment it discovered many phishing sites that were not yet reported by common engines [3]. PhishIntention extended this idea in 2022 by checking both visual brand intention and whether the page actually tries to collect credentials; the authors reported substantially fewer false alerts than reference-based baselines while maintaining similar detection of real phishing pages [8]. These systems show a shift from only asking “is this URL suspicious?” to asking “what brand is being impersonated and what is the page trying to make the user do?”

Email and language-based detection. Email remains important because it provides the first contact for many scams. Butt et al., including researchers from Pakistan, evaluated SVM, Naive Bayes and LSTM approaches for phishing-email classification and reported strong performance on their selected datasets [5]. However, survey work warns that accuracy on a fixed dataset is not enough: real campaigns change vocabulary, sender infrastructure and layout, and user-facing systems also need usable explanations [4].

Financial-fraud detection. Recent fraud research increasingly models normal customer behaviour instead of relying only on fixed rules. Vanini et al. combined anomaly detection, economic triage and risk modelling on real payment data, emphasizing that a detector should reduce financial loss, not simply maximize a machine-learning metric [9]. This is especially relevant to banks because false positives create investigation cost and customer frustration.

4. Pakistan-Focused Discussion

Pakistan has a strong reason to study phishing together with financial fraud rather than as separate topics. Abbas and Arif surveyed 384 banking customers and found that phishing, hacking and identity theft negatively affected adoption of e-banking services in Pakistan [11]. This means the impact is not limited to the immediate loss from a single scam: repeated fraud and fear of fraud can reduce trust in digital banking itself.

The regulatory response also shows that social engineering is treated as a real banking risk. In 2021, the State Bank of Pakistan (SBP) instructed banks and microfinance banks to collect information about attempted and successful digital-banking frauds reported through call centres, including fake-call frauds, and to take rapid blocking actions when fraud is reported [12]. In 2023, SBP issued additional measures to strengthen the security of digital banking products and services and linked weak controls with customer compensation obligations [13]. National CERT Pakistan also maintains phishing-awareness material and advises users on safer online behaviour [14].

From a research perspective, however, Pakistan still appears under-represented in open technical datasets. A local phishing detector should expect English, Urdu and Roman Urdu, mixed-language messages, shortened URLs, screenshots, QR codes and highly local lures such as bank/KYC updates, parcel notices, government schemes, job offers and mobile-wallet support. Models trained mainly on English email corpora or global brand lists may miss these patterns. A practical Pakistani research agenda should therefore combine technical detection with user education and financial-sector threat sharing.

5. Open-Source Practice

One useful outcome of the review is that a strong laboratory or institutional workflow does not require only commercial products. Several open-source projects cover different stages of the problem:

- **Phishpedia:** Research code and visual brand-identification approach for phishing webpages. It is useful for studying screenshot/logo-based detection and explainable brand matching [3].
- **MISP:** Open-source threat-intelligence platform for storing, correlating and sharing indicators. The project explicitly supports financial-fraud information and can export intelligence into IDS/SIEM formats [15].
- **YARA:** Open-source pattern-matching tool that can identify suspicious files or content using analyst-written rules. It is useful when phishing delivers malware or when investigators need repeatable pattern-based triage [16].
- **Zeek / Suricata:** Open-source network monitoring and intrusion-detection tools. Zeek produces detailed HTTP, DNS, TLS and connection logs that can help reconstruct access to suspicious infrastructure; Suricata can apply signatures and IOC feeds [17].
- **PhishTank and open feeds:** Community phishing URLs are frequently used in research datasets and can support validation or enrichment, although analysts must still check freshness, duplicates and labeling quality [4].

6. Unsolved Problems

1. Local and multilingual data

Pakistan-specific labelled phishing and fraud datasets are limited. A useful dataset must represent Urdu, Roman Urdu, English, SMS, WhatsApp-style messages, fake banking pages and local payment services while protecting customer privacy.

2. Adaptation to AI-assisted scams

Attackers can rewrite content, clone brands and rapidly rotate domains. Recent multimodal and LLM-based systems are promising [10], but they also introduce cost, model drift and new adversarial surfaces. Continuous evaluation is more important than one-time accuracy.

3. Cross-channel evidence sharing

A single incident may involve a fake message, a phone number, a phishing domain, a mule account and a fraudulent transaction. Banks, telecom operators, CERTs and investigators need fast sharing of indicators and evidence, but privacy, legal process and inconsistent data formats make this difficult.

7. Practical Recommendations for Pakistan

- Build a privacy-safe national phishing/fraud dataset with English, Urdu and Roman Urdu examples and clear labeling rules.
- Use layered detection: URL/domain reputation, visual/brand checks, content models, transaction anomaly detection and human review rather than relying on one model.
- Expand structured threat sharing between banks, telecom operators, CERT/CSIRT teams and investigators using open standards and tools such as MISP.
- Keep user awareness tied to real local scam examples. Technical controls are necessary, but social engineering still depends on the victim being rushed, frightened or persuaded.
- Evaluate models using false-positive cost, detection delay and financial loss prevented, not accuracy alone.

8. Conclusion

Research since 2020 shows clear progress in phishing and online-fraud detection. The field has moved from static indicators toward deep learning, visual brand recognition, behavioural analysis and multimodal models. The strongest direction is not a single algorithm but a layered workflow: detect suspicious communication, identify the impersonated brand or intent, monitor user and transaction behaviour, share indicators, and keep a human investigator in the loop. For Pakistan, the most important research opportunity is to make these methods local: local languages, local banking and wallet brands, local user behaviour and local reporting channels. Open-source tools make this realistic for universities, CERT teams and digital-forensics labs, but the quality of data and cooperation between organizations will decide how effective the final system becomes.

References (IEEE style)

- [1] A. Arshad, A. U. Rehman, S. Javaid, T. M. Ali, J. A. Sheikh, and M. Azeem, "A Systematic Literature Review on Phishing and Anti-Phishing Techniques," arXiv:2104.01255, 2021.
- [2] M. Bitaab et al., "Scam Pandemic: How Attackers Exploit Public Fear through Phishing," arXiv:2103.12843, 2021.

- [3] Y. Lin et al., "Phishpedia: A Hybrid Deep Learning Based Approach to Visually Identify Phishing Webpages," in 30th USENIX Security Symposium, 2021, pp. 3793-3810.
- [4] C. Catal, G. Giray, B. Tekinerdogan, S. Kumar, and S. Shukla, "Applications of deep learning for phishing detection: a systematic literature review," Knowledge and Information Systems, vol. 64, no. 6, pp. 1457-1500, 2022, doi: 10.1007/s10115-022-01672-x.
- [5] U. A. Butt, R. Amin, H. Aldabbas, S. Mohan, B. Alouffi, and A. Ahmadian, "Cloud-based email phishing attack using machine and deep learning algorithm," Complex & Intelligent Systems, vol. 9, pp. 3043-3070, 2023 (published online 2022), doi: 10.1007/s40747-022-00760-3.
- [6] A. Ali et al., "Financial Fraud Detection Based on Machine Learning: A Systematic Literature Review," Applied Sciences, vol. 12, no. 19, art. 9637, 2022, doi: 10.3390/app12199637.
- [7] W. Hilal, S. A. Gadsden, and J. Yawney, "Financial Fraud: A Review of Anomaly Detection Techniques and Recent Advances," Expert Systems with Applications, vol. 193, art. 116429, 2022, doi: 10.1016/j.eswa.2021.116429.
- [8] R. Liu et al., "Inferring Phishing Intention via Webpage Appearance and Dynamics: A Deep Vision Based Approach," in 31st USENIX Security Symposium, 2022, pp. 1633-1650.
- [9] P. Vanini, S. Rossi, E. Zvizdic, and T. Domenig, "Online payment fraud: from anomaly detection to risk management," Financial Innovation, vol. 9, art. 66, 2023, doi: 10.1186/s40854-023-00470-w.
- [10] J. Lee, P. Lim, B. Hooi, and D. M. Divakaran, "Multimodal Large Language Models for Phishing Webpage Detection and Identification," arXiv:2408.05941, 2024.
- [11] T. Abbas and K. Arif, "End-Users' Perception of Cybercrimes Towards E-banking Adoption and Retention," Journal of Independent Studies and Research - Computing, vol. 21, no. 1, 2023, doi: 10.31645/JISRC.23.21.1.10.
- [12] State Bank of Pakistan, "Collection of Digital Banking Frauds/Attempted Frauds Information through Call Centers," Circular, Feb. 18, 2021.
- [13] State Bank of Pakistan, "Measures to Enhance Security of Digital Banking Products and Services," BPRD Circular No. 04 of 2023, Apr. 14, 2023.
- [14] National CERT Pakistan, "Stay Secure Against Phishing," ABC Awareness Program, accessed Aug. 2026.
- [15] MISP Project, "Open Source Threat Intelligence and Sharing Platform," accessed Aug. 2026.
- [16] VirusTotal, "YARA - The pattern matching swiss knife," open-source project documentation, accessed Aug. 2026.
- [17] Zeek Project, "The Open Source Network Security Monitoring Tool," accessed Aug. 2026.

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